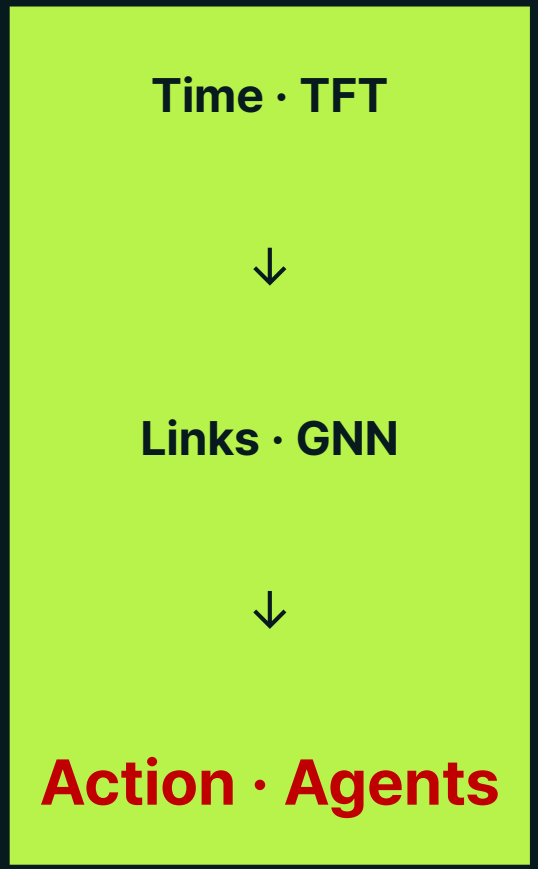
TFT · GNN · Agents
Modern Model Structures

See the problem each design solves, not the new name



Financial ML · Week 15

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Data and forecast shape** Tables, series, graphs; point vs. quantile
02	**Core concepts**	**How the three work** TFT inputs, message passing, agents
03	**Cases**	**Origins and real links** TFT paper, supply chain shocks, tech layers
04	**Summary**	**Versus baselines** Check Q&A

A good system need not use all three

01

Basics

Data structure and forecast forms

Data shape determines model structure

The same financial data fits different models by layout

TABLE

Tabular

- Each row is an independent sample.

TIME SERIES

Time series

- Order matters; the past shapes the future.

GRAPH

Graph

- Links between nodes carry meaning.

TFT is built for time series, GNN for graphs

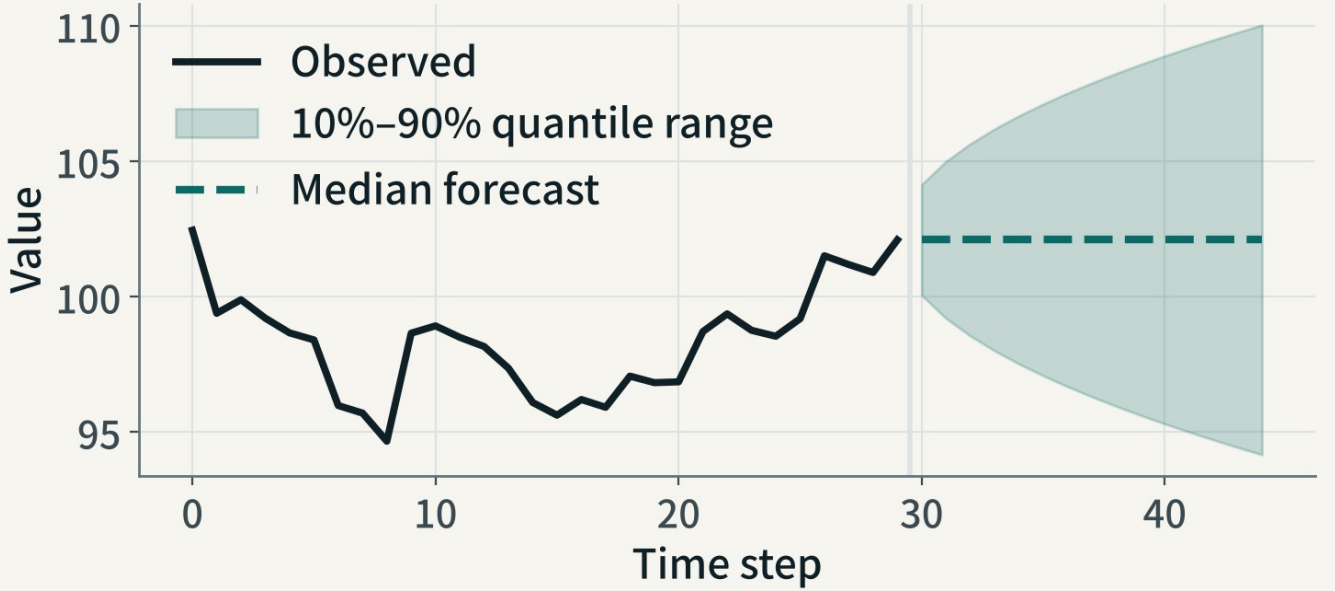
Misread data shape, and validation design fails

A quantile forecast gives a range, not one number

TFT can express uncertainty with quantile forecasts

Vs. point forecast

- A point forecast gives the median only.
- A quantile forecast also gives 10th and 90th percentiles.



※ Example data — the range widens further ahead

Wider range, more uncertainty

In risk management, the range often beats the median

02

How the three work

Source concepts: TFT, weighted average, GNN, agents

The three are not rival models of one kind

TFT forecasts time series, GNN learns on relations, agents carry out tasks

Target	Extra info used	Good question
TFT	**Time context, known future inputs**	**Demand·risk at many horizons?**
GNN	**Firm·account·trade links**	**What do neighbors tell me?**
AI agent	**Tool results, iteration**	**How to finish research and reports?**

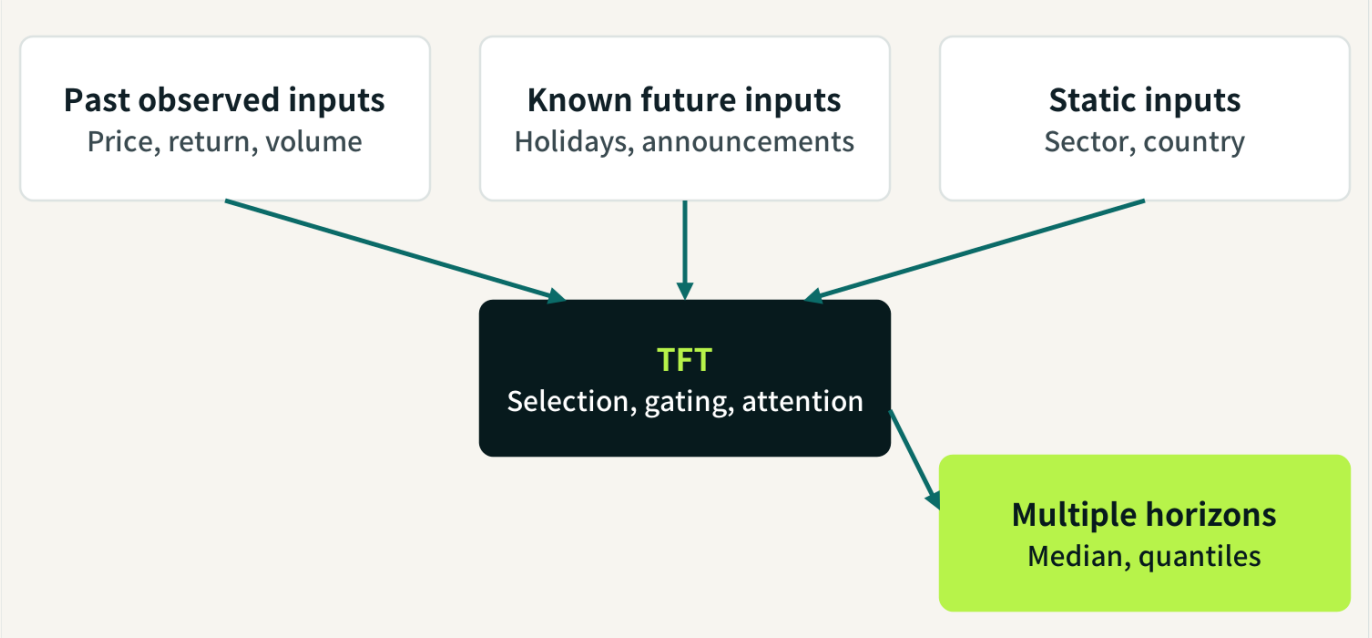
‘Modern models’ here means the three in the repository

TFT separates past observations, known future inputs, static traits

TFT (Temporal Fusion Transformer) — forecasting tomorrow's store sales

Key: separation

- Holidays are known ahead.
- Earnings dates are known; figures are not.
- Sector, country: as-of data.



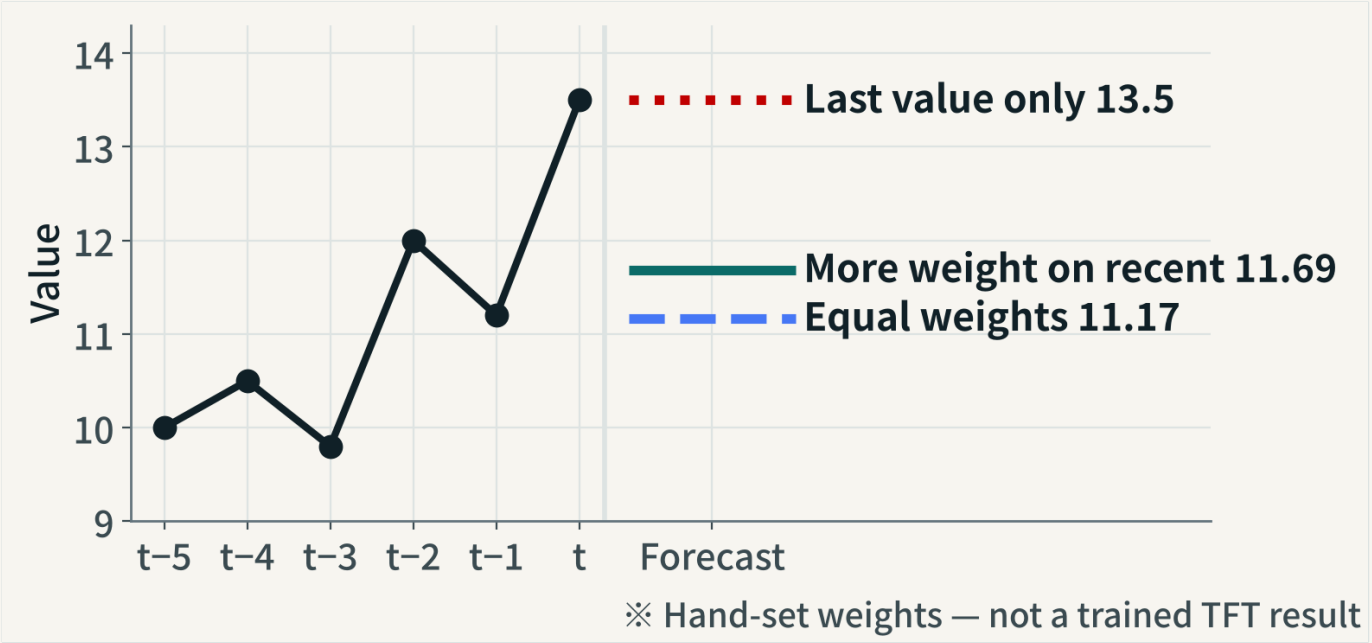
TFT combines variable selection, gating, recurrence and attention for multi-step output

The same past gives different forecasts by where weight goes

Hand calculation: different weights on source values 10, 10.5, 9.8, 12, 11.2, 13.5

Results

- Equal average: about 11.17.
- Weights 1–6: about 11.69.
- Last value only: 13.5.



Attention intuition only

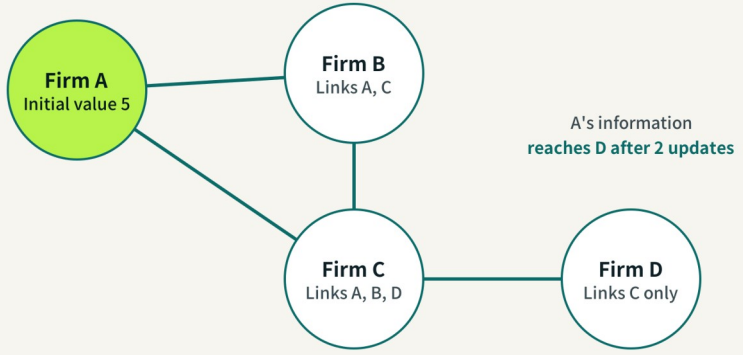
A weighted average is not TFT — these weights were set by hand

A GNN updates itself by gathering neighbor information

Nodes: firms, accounts; edges: supply·loan·trade links — all update at once from last round

Source toy-model update

$$h_v^{new} = 0.5\, h_v^{old} + 0.5 \times \text{mean}_{u \in N(v)}\, h_u^{old}$$



	A	B	C	D
Start	5	0	0	0
1 update	2.500	1.250	0.833	0
2 updates	1.771	1.458	1.042	0.417

Message-passing rounds set how many hops away a node can see

Toy message passing is not full GNN training

Fixed neighbor averaging: no learnable weights, labels or loss

Real design

- Set edge direction, type, timing, aggregation, parameters.

Oversmoothing

- Too many layers make nodes look alike.

Round ≠ time

- A round is not a day or real shock-propagation time.

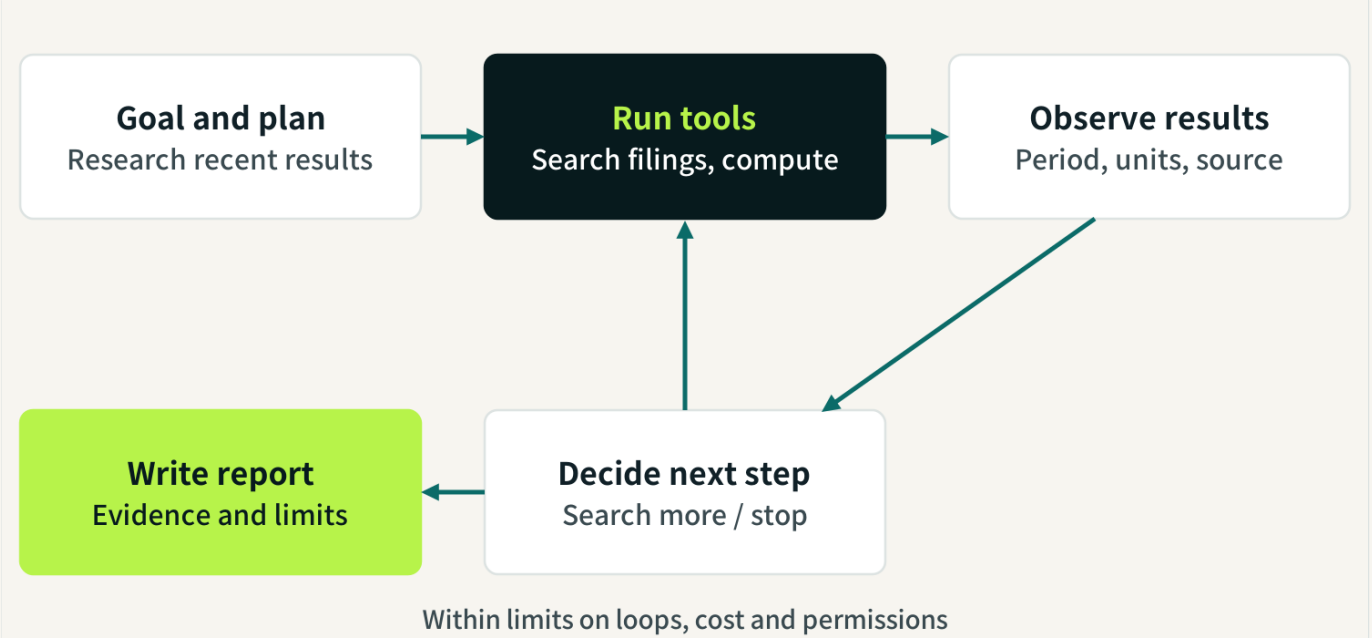
Graph links do not imply causation

An AI agent picks its next action from tool results

Goal: ‘Research firm A's latest quarter and make a table’

Key

- Find filings, read tables, align units, compute.
- Search again if needed, then report.



Classify by flow, not product name

Chatbots use tools too; check if each result shapes the next step

Expected value rises when accuracy gains exceed tool cost

Source example: +10 if right, −20 if wrong, 1 per tool call

Expected value

Memory only: $0.6(10) - 0.4(20) = -2$

One retrieval: $0.95(10) - 0.05(20) - 1 = 7.5$

Assumptions

- Accuracies 0.60 and 0.95 are illustrative.

Source's rules

- Set tool scope and cost limits.
- Humans review before live trades.

Agents can also misread or pick the wrong tool

The three can link up, each on a different layer

Time → relations → workflow

TFT

Time

Patterns in each series

GNN

Links

Links between firms

AI agent

Workflow

Collect, run, report

A good system need not use all three

Weigh validated gains over simple baselines against complexity

03

Lessons from cases

Origins of each structure and real-world links

TFT was proposed for interpretable multi-horizon forecasts

Lim et al. (2021, International Journal of Forecasting): Google researchers

The problem

- Mixing input types to forecast many horizons at once.

Interpretation

- Variable importance and attention weights show inputs used.

Caution

- Revalidate the paper's results OOS on your own data.

Interpretability is not causal explanation

Supply chain links are paths for shocks

2021 auto chip shortage: parts disruption cut car output

What happened

- When chip supply stalled, many automakers cut output.

As a graph

- Chipmakers and automakers link by supply edges.

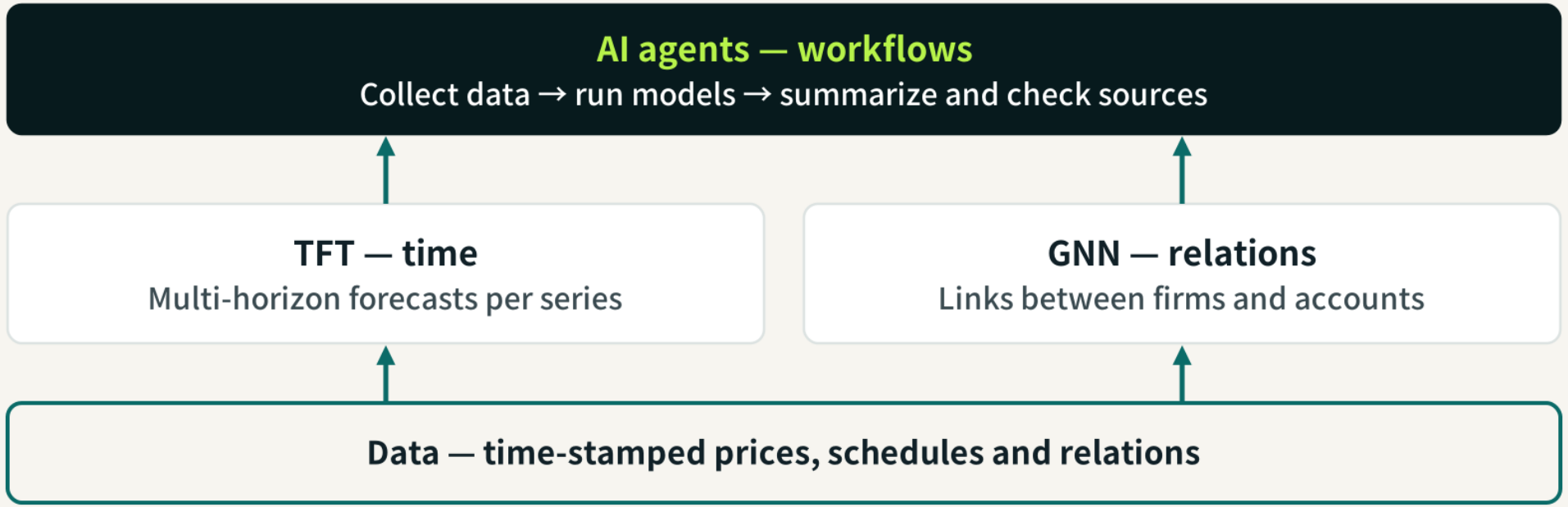
Caution

- Use point-in-time links; links do not prove causation.

Relational data shows what table rows alone cannot

The three technologies stack in layers on data

Models handle time and relations; agents run workflows



Wrong timing below makes every layer above wrong

Compare gains over simple baselines with operating complexity

04

Summary

Close the week with key Q&A

Judge by the problem solved and validation, not the name

One question, one answer for this week

Check question	Answer
Is any attention model a TFT?	**No. TFT is a specific structure combining several parts.**
Does message passing prove causation?	**No. It processes info over given links.**
Weighted-average lab = TFT training?	**No. It builds intuition with human-set weights.**
Can we trust a tool-using agent?	**It can misread or misselect; it needs limits and review.**

Compare all three with simple baselines on equal terms

TIME. RELATIONS. ACTIONS.

EACH HAS ITS SHAPE.

TEST AGAINST SIMPLE.

A new structure matters only
if it beats baselines.